Slope and Rate of Change — Diagnosing Inverted and Sign-Reversed Calculations

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

1. 2	4. -18	7. -1.5
2. -2	5. -3	8. 5
3. $\frac{1}{2}$	6. 2	

Common wrong answers

1. *If they got 0.5: inverted the ratio: run over rise instead of rise over run*
2. *If they got 2: subtracted the y-values in one order and the x-values in the other, reversing the sign*
3. *If they got 2: divided the change in x by the change in y, inverting the slope*
4. *If they got 18: subtracted the smaller reading from the larger one, losing the sign of the change*
5. *If they got 3: subtracted the levels in one order and the times in the other*
6. *If they got -2: took the y-difference from the first point to the second but the x-difference the other way*
7. *If they got -0.67: inverted the ratio, dividing the change in time by the change in height*

1. Slope through $(2, 3)$ and $(6, 11)$.

Step 1: label the points in one fixed order: $(x_1, y_1) = (2, 3)$ and $(x_2, y_2) = (6, 11)$. Step 2: rise first (it belongs on top): $y_2 - y_1 = 11 - 3 = 8$. Step 3: run underneath, subtracted in the SAME order: $x_2 - x_1 = 6 - 2 = 4$. Step 4: $m = \dfrac{8}{4}$, so $\boxed{m = 2}$. Sanity check: from $(2, 3)$ the line climbs 8 while moving 4 right, so a slope greater than 1 is right.
Watch for: 0.5 — the run divided by the rise.

2. Find and fix: Devi's slope through $(-3, 5)$ and $(2, -5)$.

Step 1: read her set-up as an order question. On top she wrote $5-(-5)$: that is $y_1 - y_2$, first point minus second. Underneath she wrote $2 - (-3)$: that is $x_2 - x_1$, second point minus first. Step 2: name the error — the two differences were taken in *opposite* orders, which multiplies the answer by -1. That is a sign reversal, not an arithmetic slip; her 10 and her 5 are both correct numbers. Step 3: redo it in one order, second minus first throughout: $m = \dfrac{-5-5}{2-(-3)} = \dfrac{-10}{5}$. Step 4: $\boxed{m = -2}$. Check the direction: y falls from 5 to -5 while x increases, so the slope must be negative — Devi's $+2$ could have been rejected before any arithmetic.

3. Find and fix: Mika's slope through $(1,4)$ and $(5,6)$.

Step 1: identify what each of his differences measures. His numerator $5-1$ is the change in x, and his denominator $6-4$ is the change in y: the fraction is upside down. Step 2: name the error — inverted ratio (run over rise). His answer is the *reciprocal* of the true slope, which is why it is 2 instead of a number below 1. Step 3: rebuild it as rise over run: $m = \dfrac{6-4}{5-1} = \dfrac{2}{4}$. Step 4: simplify: $\boxed{m = \dfrac{1}{2}}$. Check: moving 4 right lifts the line only 2, so the line is shallow and the slope must be less than 1.

4. Change in the tank's level from hour 0 to hour 6.

Step 1: read both values off the table: at hour 0 the level is 46 litres, and at hour 6 it is 28 litres. Step 2: a change is always *later minus earlier* — the order carries the meaning, so it is $28-46$, not $46-28$. Step 3: $28-46 = \boxed{-18}$ litres. Step 4: check the sign against the story: the tank is draining, so the change must be negative.
Watch for: 18 — the same size, but it describes a tank filling up.

5. Rate of change of the tank's level, hour 0 to hour 6.

Step 1: a rate of change is the slope of the table: change in litres over change in hours, both taken later-minus-earlier. Step 2: numerator (from problem 4): $28-46=-18$ litres. Step 3: denominator, same order: $6-0=6$ hours. Step 4: $\dfrac{-18}{6} = \boxed{-3}$, that is, 3 litres lost per hour. Step 5: check with a single step of the table: from hour 0 to hour 2 the level drops from 46 to 40, which is 6 litres in 2 hours — the same -3 per hour.
Watch for: 3 — correct size, wrong sign, and it claims the tank is filling.

6. Find and fix: Ty's slope through $(5,9)$ and $(1,1)$.

Step 1: check the order of each difference. Ty's numerator $1-9$ runs from the first point's y to the second point's y; his denominator $5-1$ runs the other way, second to first. Step 2: name the error — mixed order, so his answer carries the wrong sign. Step 3: fix the order and keep it: taking $(5,9)$ as point 1, $m = \dfrac{1-9}{1-5} = \dfrac{-8}{-4}$. Step 4: two negatives divide to a positive: $\boxed{m=2}$. Step 5: direction — a positive slope means the line **rises** from left to right. Check it against the points: as x grows from 1 to 5, y grows from 1 to 9.
Watch for: -2 — Ty's answer; the size is right, so only the sign diagnosis separates a correct paper from his.

7. Candle: 18 cm at 3 minutes, 6 cm at 11 minutes. Diagnose Jo's rate and find the correct one.

Step 1: read Jo's fraction. She put the time difference $11-3$ on top and the height difference $6-18$ underneath — minutes per centimetre, the reciprocal of what was asked. Her orders were consistent, so her only error is the inversion. Step 2: set it up as height change over time change, later minus earlier in both: $\dfrac{6-18}{11-3}$. Step 3: numerator $6-18=-12$ cm; denominator $11-3=8$ minutes. Step 4: $\dfrac{-12}{8} = \boxed{-1.5}$, that is, the candle loses 1.5 cm each minute. Step 5: check plausibility: in 8 minutes it should lose $8\times 1.5 = 12$ cm, and it does (18 down to 6).
Watch for: -0.67 — Jo's inverted value; note it is between -1 and 0, far too slow for a candle that lost 12 cm in 8 minutes.

8. Challenge: the line through $(a, 4)$ and $(7, -2)$ has slope -3. Diagnose Priya's set-up and solve for a.

Step 1: diagnose. Priya wrote $\frac{7-a}{-2-4}$: the x-difference is on top and the y-difference underneath, so her equation says run over rise $= -3$. It is the inverted slope, set equal to the true slope. Step 2: write the correct equation, rise over run, both differences second minus first: $\frac{-2-4}{7-a} = -3$. Step 3: simplify the numerator: $\frac{-6}{7-a} = -3$. Step 4: multiply both sides by $(7-a)$: $-6 = -3(7-a) = -21 + 3a$. Step 5: add 21 to both sides: $15 = 3a$, so $\boxed{a = 5}$. Step 6: check by recomputing the slope with $a = 5$: through $(5, 4)$ and $(7, -2)$, $m = \frac{-2-4}{7-5} = \frac{-6}{2} = -3$ ✓. Priya's -11 fails this check: through $(-11, 4)$ and $(7, -2)$ the slope is $\frac{-6}{18} = -\frac{1}{3}$, the reciprocal of -3 — the signature of an inverted set-up.